Project Completion (Close-out) Report

Project: Integration of DIA Price Oracles on Cardano
Catalyst Fund: Fund 14 — Cardano Use Cases: Concepts
Project number: 1400073
Project Manager: Dillon Hanson (dillon.hanson@diadata.org)
Start date: 2026-04-14
Completion date: 2026-08-10

Project Completion Video: https://youtu.be/iPOC-Ojb5k8
Repository: https://github.com/diadata-org/dia-cardano-oracle (MIT License)
Permanent link to this document:
https://github.com/diadata-org/dia-cardano-oracle/blob/catalyst-m4-closeout/docs/milestones/close-out-report.pdf

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1. Deliverables

DIA's push-oracle protocol was ported to Cardano (Aiken, Plutus V3), deployed on Mainnet, fed continuously by a purpose-built Cardano feeder, monitored end to end, and made consumable by any Cardano dApp through a read-only indexer and an example consumer contract.

On-chain evidence:

Artifact	Reference
Live mainnet Pair (oracle feed), client `client-test-01`	Pair policy id `b1b933a7b08ebdee6d957b4ae3d027ac4a13f9d319dbd8a2b95e052f` — indexer README, published contract addresses
Headline mainnet oracle-update transaction	`01315fb64b120a6852722b28f880dfb05f75b78ba9374b16e3771e3c2560fcea`
Full mainnet deployment chain-walk (Milestone 1)	`milestone-1-poa.md`
Sustained mainnet reliability window (Milestone 4)	40 confirmed on-chain updates, 2026-07-13 09:45 → 2026-07-14 18:01 UTC — mainnet evidence pack
Preview deployment — 16 distinct live feeds	preview evidence pack

Off-chain evidence:

Component	Location
Aiken contracts (7 validators) + unit tests	`contracts/aiken/` — 167/167 tests passing
Off-chain CLI (protocol/client bootstrap, maintenance)	`offchain/cli/` — 59/59 tests passing
Feeder daemon (DIA → Cardano pipeline)	`offchain/feeder/` — 790/790 tests passing
Indexer (consumer-facing query service)	`offchain/indexer/` — 67/67 tests passing
Monitoring (Prometheus + Grafana + Alertmanager)	`offchain/feeder/monitoring/`, dashboards guide
Architecture + security documentation	architecture, security notes

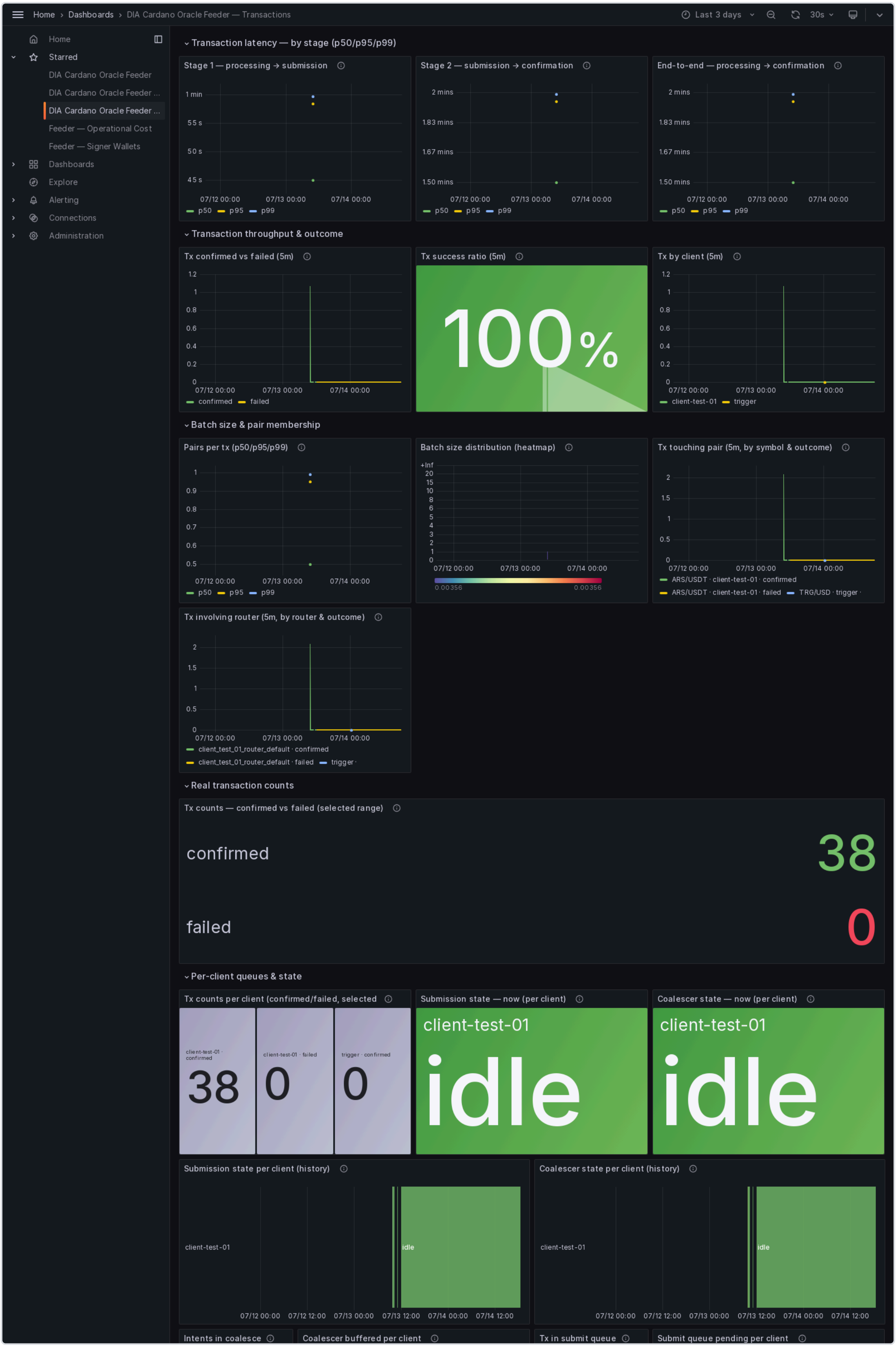
Hosting: the repository is public on GitHub (MIT License); no separate hosted service is required to run any component — every service (feeder, indexer, monitoring) is a Docker image + `docker-compose` a third party can run themselves against their own chain-provider key.

Testing: all four test suites (Aiken, CLI, feeder, indexer) were re-run and verified passing during this milestone's evidence review — 1,083 automated tests in total. See the per-suite counts above and the raw console output in each evidence pack's `tests/` directory.

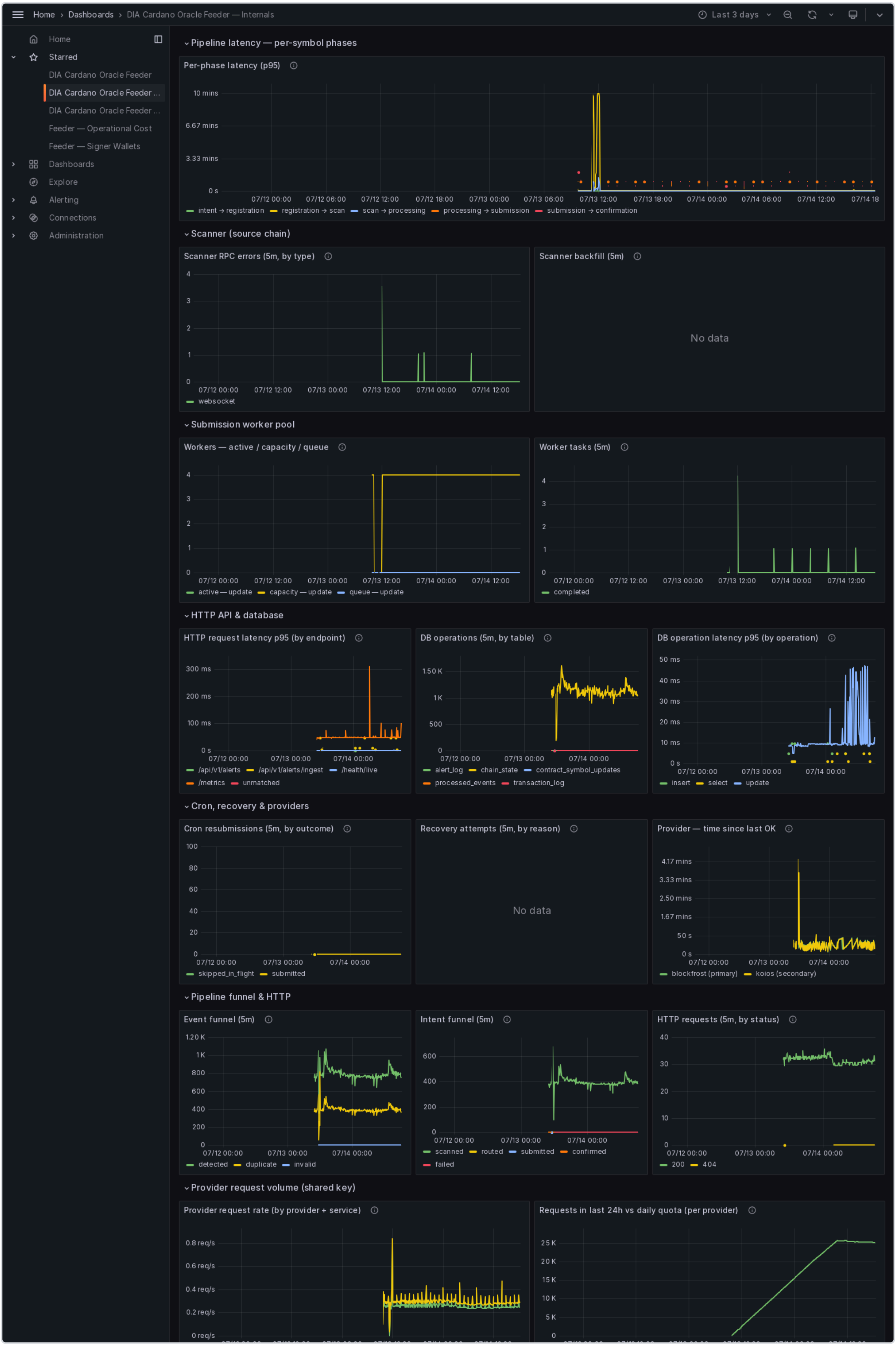
Visual evidence

The five Grafana dashboards the system ships, rendered live over the Mainnet run window (2026-07-13 → 2026-07-14). Full-resolution originals: `dashboards/`.

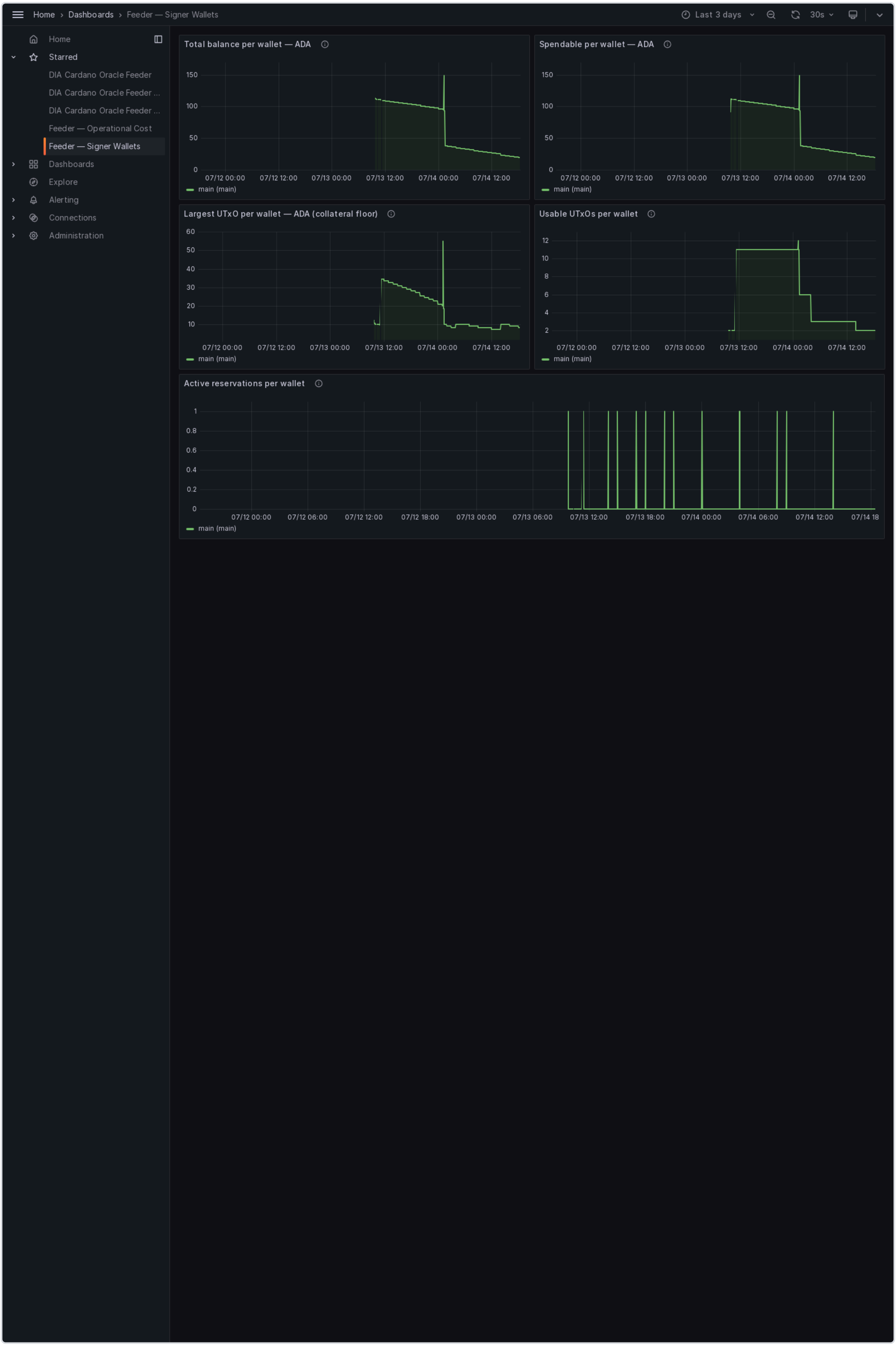
Overview — per-feed health: is each price feed alive, fresh, accurate and funded.



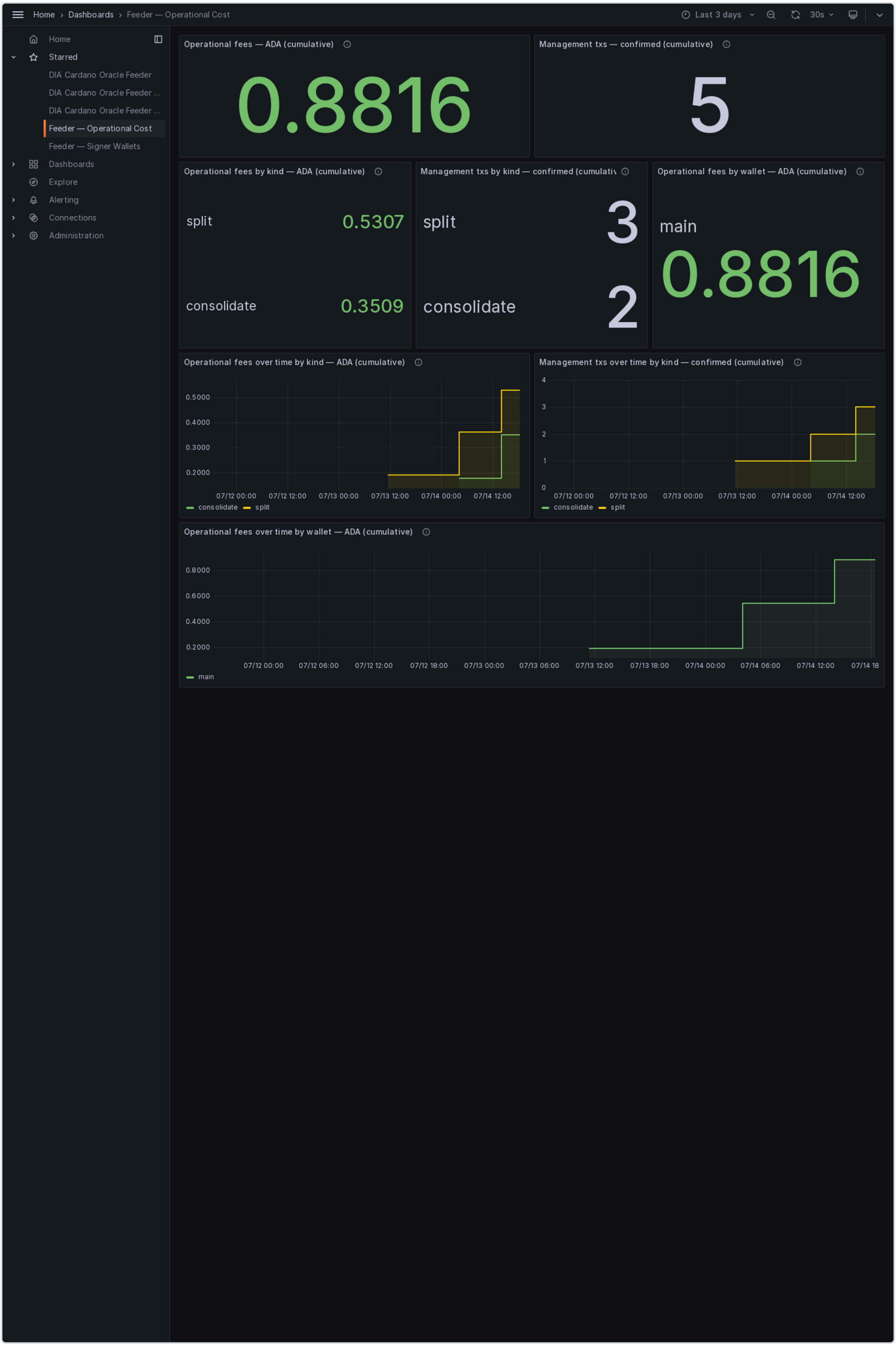
Transactions — every oracle update published to Cardano, its confirmation and its outcome.



Internals — pipeline latency, scanner, worker pools, database and provider health.



Signer Wallets — the multi-wallet signing pool that publishes the updates.



Operational Cost — the ADA cost of running the system, by transaction kind and signing wallet.

2. Usage

The intended users are **Cardano dApp developers who need a DIA price feed**. They interact with the system exclusively through the indexer's read-only HTTP API — no CLI, no admin access, no on-chain transaction of their own required to read a price:

1. Query `GET /v1/pairs/{symbol}` for the latest value and the exact UTxO to reference.
2. Include that Pair UTxO as a reference input in their own transaction and authenticate it by its Pair NFT (`GET /v1/protocol/fees` for the cost of keeping a subscribed feed updated).
3. Optionally consume through the published example consumer contract as a working reference implementation.

On-chain activity to date, independently verifiable on Cardanoscan:

Measure	Value
Confirmed oracle-update transactions on Mainnet (window)	40
Live client on Mainnet	1 (`client-test-01`)
Continuously published Mainnet feed	`ARS/USDT`
Distinct feeds live and queryable on Preview	16
Mainnet live since	2026-06-16

No integrations are confirmed yet, as the oracle has just gone live. DIA is in active outreach and preliminary discussions with a number of Cardano-ecosystem protocols, including Liqwid Finance, Indigo Protocol, Djed Alliance, Fluid Tokens, Surf Lending, Ascend, Dano Finance, Strike Finance, Minswap, Sundaeswap, Muesliswap, WingRiders, and VyFinance. DIA will continue connecting with partner prospects across the Cardano ecosystem and is now beginning support in the live production environment. Because no third-party dApp is consuming the feed yet, no external user-count is reported here; the transaction figures above are the protocol's own published oracle updates.

3. Impact

Measurable, verified results as of this report:

- **100% observed broadcast-publication success** on the live Mainnet feed (`ARS/USDT`) over a ~32.27 h sustained window: 40 confirmed updates, 0 real on-chain transaction failures, and 0 chain reorgs.
- **32.3 s median time to on-chain confirmation** across those 40 updates (average 36.9 s, fastest 19.2 s, slowest 87.7 s) — recomputable from the transaction log shipped in the mainnet evidence pack.
- **0.90 ADA average cost per published oracle update**, measured from the 32.24 ADA of Cardano fees paid across the sampled updates — about 645 ADA per year-round feed at the hourly cadence used here. This is a measured operating cost, not an estimate.
- **99.78% freshness compliance** against a strict one-hour ceiling. This is a deliberately conservative timing figure: it counts ordinary scheduling and block-confirmation latency as staleness, and is not a measure of downtime. The method is documented in the mainnet evidence pack.
- **16 distinct price feeds** deployed and live-queryable on Preview, demonstrating the architecture's capacity beyond a single feed.
- **1,083 automated tests passing** across contracts, CLI, feeder, and indexer.
- Cardano previously had no DIA oracle integration; this closes that gap and gives any Cardano dApp a documented path to any of DIA's 2,500+ price feeds and 10,000+ real-world-asset feeds.

No TVL or ecosystem-adoption figures are reported yet, as the oracle has just gone live. DIA is now initiating the onboarding process with protocols across the ecosystem.

4. Sustainability

Ongoing project — maintenance and roadmap:

- The oracle, feeder, and indexer are designed to run continuously; the feeder already carries the operational tooling (automated fee settlement, wallet maintenance, alerting) needed for unattended production operation.
- Requesting a new feed is a standing, documented process — see indexer README — requesting a new feed — not a one-off integration effort.
- The protocol carries its own on-chain revenue mechanism: each client holds a prepaid balance in an on-chain Receiver and is charged the published protocol fee per update, queryable at `GET /v1/protocol/fees`.
- DIA will fund the ongoing infrastructure and maintenance costs for the deployed price oracles. Costs tied to custom NAV oracle requests, or additional features outside the scope of the original grant, will be discussed and negotiated directly with the requesting protocol, or proposed as a follow-up Catalyst submission.

Forking / permanent availability (in case the project is not continued):

- The repository is public, MIT-licensed, and pinned per milestone by commit — anyone can fork, redeploy, or extend it independently of DIA or Protofire.
- The compiled contract artifact (`contracts/aiken/plutus.json`) is committed, so a fork does not depend on rebuilding from source or on any hosted service remaining available.
- All on-chain state (Config, Receiver, Pair UTxOs) is public and independently readable by the indexer from any chain-provider key — no centralized dependency on Protofire's or DIA's own infrastructure is required to read a published feed.

Pointers

- Project Completion Video: https://youtu.be/iPOC-Ojb5k8
- Milestone 4 Proof of Achievement: `milestone-4-poa.md`
- Milestone 1–3 Proofs of Achievement: 1, 2, 3
- Protocol architecture: `cardano-oracle-architecture.md`
- License: `LICENSE` (MIT)